

Homework 12

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1. Let (e_j) be an orthonormal sequence in a Hilbert space H . Show that if $x = \sum_{j=1}^{\infty} a_j e_j$ and $y = \sum_{j=1}^{\infty} b_j e_j$, then

$$\langle x, y \rangle = \sum_{j=1}^{\infty} a_j \overline{b_j}.$$

Proof. We have

$$\begin{aligned} \langle x, y \rangle &= \left\langle \sum_{i=1}^{\infty} a_i e_i, \sum_{j=1}^{\infty} b_j e_j \right\rangle \\ &= \sum_{i=1}^{\infty} a_i \left\langle e_i, \sum_{j=1}^{\infty} b_j e_j \right\rangle \\ &= \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_i \overline{b_j} \langle e_i, e_j \rangle \\ &= \sum_{i=1}^{\infty} a_i \overline{b_i} \langle e_i, e_i \rangle \\ &= \sum_{i=1}^{\infty} a_i \overline{b_i} \end{aligned}$$

as desired, where the fourth equality comes from the orthogonality of e_i with e_j when $i \neq j$, and the final equality comes from the normality of each e_i . $\square$

2. Let (e_j) be an orthonormal sequence in a Hilbert space H , and let $U = \text{span}\{e_1, e_2, \dots\}$. Show that for any $x \in H$, the vector $y = \sum_{j=1}^{\infty} \langle x, e_j \rangle e_j$ satisfies $y \in \overline{U}$ and $x - y \in (\overline{U})^{\perp}$.

Proof. Since U is a subspace, we can use the fact that $U^{\perp\perp}$ is the smallest subspace containing U (proved in a previous homework), and so $U^{\perp\perp} = \overline{U}$. In fact, since $U^{\perp}$ is closed, we have $\overline{U}^{\perp} = U^{\perp\perp\perp} = U^{\perp}$. We thus begin by showing that $x - y \in \overline{U}^{\perp}$ by showing that $x - y \in U^{\perp}$.

Indeed, we take the inner product of $x - y$ with a general member of U to get

$$\begin{aligned}
\left\langle x - y, \sum_{j=1}^{\infty} a_j e_j \right\rangle &= \left\langle x - \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i, \sum_{j=1}^{\infty} a_j e_j \right\rangle \\
&= \left\langle x, \sum_{j=1}^{\infty} a_j e_j \right\rangle - \left\langle \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i, \sum_{j=1}^{\infty} a_j e_j \right\rangle \\
&= \sum_{j=1}^{\infty} \overline{a_j} \langle x, e_j \rangle - \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \langle x, e_i \rangle \overline{a_j} \langle e_i, e_j \rangle \\
&= \sum_{j=1}^{\infty} \overline{a_j} \langle x, e_j \rangle - \sum_{i=1}^{\infty} \langle x, e_i \rangle \overline{a_i} \langle e_i, e_i \rangle \\
&= \sum_{j=1}^{\infty} \overline{a_j} \langle x, e_j \rangle - \sum_{i=1}^{\infty} \langle x, e_i \rangle \overline{a_i} \\
&= 0,
\end{aligned}$$

where the fourth equality comes from the orthogonality of e_i and e_j when $i \neq j$, and the fifth equality comes from the normality of each e_i . Since our choice of element from U was arbitrary, $x - y \in U^\perp = \overline{U}^\perp$. $\square$

3. Prove the following theorem.

Theorem 1. Let H be a Hilbert space, and $M \subset H$ be an orthonormal set. Then the following statements are equivalent:

- (1) M is total.
- (2) The only $x \in H$ that is orthogonal to all vectors in M is 0.
- (3) $\sum_{e \in M} |\langle x, e \rangle|^2 = \|x\|^2$ for all $x \in H$.

Proof. ((1) $\implies$ (2)) If M is total, then $H = \overline{\text{span}(M)} = \text{span}(M)^{\perp\perp}$. Also, if $y \in M^\perp$, then $y \in \text{span}(M)^\perp$ since

$$\left\langle y, \sum_{m \in M} a_m m \right\rangle = \sum_{m \in M} \overline{a_m} \langle y, m \rangle = 0$$

since y is orthogonal to each m , noting that we are considering such linear combinations when $a_m = 0$ for all but finitely many m , since that set is exactly the span of M . In other words, given some $y \in M^\perp$, every possible choice of $x \in H$ satisfies $\langle x, y \rangle = 0$ when y is orthogonal to all vectors in M . Thus y is orthogonal to every vector in H , and the only vector which has that property is 0, so $y = 0$.

((2) $\implies$ (1)) Since the set $M^\perp = \{0\}$ is closed, we have that

$$\begin{aligned}
H &= M^\perp \oplus M^{\perp\perp} \\
&= M^\perp \oplus \overline{\text{span}(M)} \\
&= \{0\} \oplus \overline{\text{span}(M)} \\
&= \overline{\text{span}(M)},
\end{aligned}$$

so M is total as desired.

((1) $\implies$ (3)) Assume M is total. If $x \in \text{span}(M)$, then for some finite $F \subset M$,

$$\begin{aligned}
\|x\|^2 &= \left\| \sum_{e \in F} \langle x, e \rangle e \right\|^2 \\
&= \left\| \sum_{e \in M} \langle x, e \rangle e \right\|^2 \quad (\text{since } x \text{ is orthogonal to every } e \in M \setminus F) \\
&= \left\langle \sum_{e \in M} \langle x, e \rangle e, \sum_{f \in M} \langle x, f \rangle f \right\rangle \\
&= \sum_{e \in M} \sum_{f \in M} \langle x, e \rangle \overline{\langle x, f \rangle} \langle e, f \rangle \\
&= \sum_{e \in M} \langle x, e \rangle \overline{\langle x, e \rangle} \langle e, e \rangle \quad (\text{since } e \perp f \text{ when } e \neq f) \\
&= \sum_{e \in M} \langle x, e \rangle \overline{\langle x, e \rangle} \\
&= \sum_{e \in M} |\langle x, e \rangle|^2
\end{aligned}$$

as desired.

Now if $x \in H \setminus \text{span}(M) = \overline{\text{span}(M)} \setminus \text{span}(M)$, then

((3) $\implies$ (1))

□